



National
Qualifications
2024

X819/76/11

Design and Manufacture

WEDNESDAY, 1 MAY

1:00 PM – 3:15 PM

Total marks — 80

SECTION 1 — 25 marks

Attempt ALL questions.

SECTION 2 — 55 marks

Attempt ALL questions.

Write your answers clearly in the answer booklet provided. In the answer booklet you must clearly identify the question number you are attempting.

Use **blue** or **black** ink.

Before leaving the examination room you must give your answer booklet to the Invigilator; if you do not, you may lose all the marks for this paper.



Downloaded free from <https://sqa.my/>

SECTION 1 — 25 marks

Attempt ALL questions

1. Two sun loungers are shown below with product information.

Luxury sun lounger

Materials

mahogany frame

textured ABS tyre

stainless steel nuts/bolts

Additional details

four adjustable positions

self-assembly (tools included)

Price – £249.99



Folding sun lounger

Materials

tubular aluminium frame

nylon netting

polypropylene footpads and
corner protectors

aluminium leg brackets

Additional details

five adjustable positions

Price – £59.99

1. (continued)

- | | |
|---|---|
| (a) Explain why the materials chosen are suitable for the sun loungers.
(You must give <i>six different explanations.</i>) | 6 |
| (b) Name three appropriate manufacturing processes used in the production of the sun loungers and explain why each one is suitable. | 6 |
| (c) Describe how function has influenced the design of the sun loungers. | 5 |
| (d) Describe how anthropometrics and physiology have influenced the design of the sun loungers. | 4 |
| (e) Explain the benefits of using computer aided design (CAD) during the design of the sun loungers. | 4 |

[Turn over

SECTION 2 — 55 marks

Attempt ALL questions

2. A children's play frame is shown below.



The slide and side panels of the play frame are rotational moulded.

- (a) Explain why rotational moulding is a suitable process for these parts. 2
- (b) State the name of a suitable material for these parts and explain why this material is appropriate. 3

It is important that products are environmentally friendly.

- (c) Outline steps that may have been taken to reduce the negative impact of products on the environment. 4

3. The success of a product relies on effective market research.

(a) Describe the key stages of **two** research techniques.

4

Different types of specification are produced for a variety of reasons.

(b) Outline the type of information that may be contained in **two** of the following specification types:

- product design specification
- performance specification
- technical specification.

4

4. During the design process, designers will use a variety of graphics and physical models.

(a) Describe how graphics could be used during the design process.

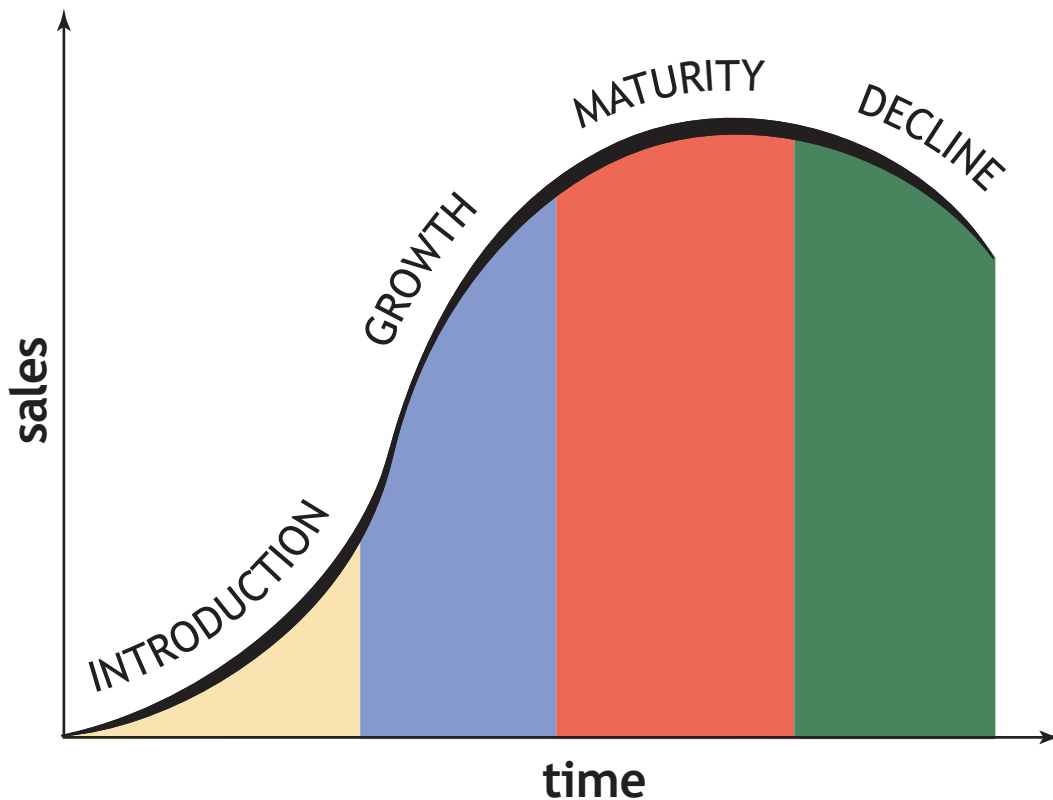
4

(b) Describe how physical models could be used during the design process.

4

[Turn over

5. A product life cycle graph is shown below.



During the introduction stage, marketing techniques are used.

(a) Describe marketing techniques that could be used to launch new products. 4

A strong brand image can increase sales.

(b) Explain why a strong brand image can lead to increased sales. 3

At the end of the product life cycle, sales will begin to decline.

(c) Explain the factors that may cause product sales to decline. 3

Copyright is a method of protecting intellectual property.

(d) Identify another method of protecting intellectual property **and** give an example of what it protects. 2

	MARKS
6. There is often a need to identify the materials used in the manufacture of products.	
(a) Describe methods that could be used to identify the materials used in a product. <i>(You should reference different materials in your answer.)</i>	4
Designers must ensure products can be assembled accurately and efficiently.	
(b) Describe how manufacturing features could be used to ensure accurate and efficient assembly.	2
Products can be manufactured using automated production methods.	
(c) Describe how automated production methods could impact the workforce and society.	4
7. Production and planning systems are used in the development and manufacture of new products.	
Explain why production and planning systems are used. <i>(You should refer to a range of different production and planning systems in your answer.)</i>	8

[END OF QUESTION PAPER]

[BLANK PAGE]

DO NOT WRITE ON THIS PAGE

Acknowledgement of copyright

- Question 1 (Luxury sun lounger) Image of 'Berkfield Sun Lounger Solid Acacia Wood.'
SQA has made every effort to trace the owners of copyright of this item and seek permissions. We are happy to discuss permission requirements and incorporate any missing acknowledgement. Please contact question.papers@sqa.org.uk.
(Folding sun lounger) Image of 'Outsunny Foldable Sun Lounger in Blue.'
SQA has made every effort to trace the owners of copyright of this item and seek permissions. We are happy to discuss permission requirements and incorporate any missing acknowledgement. Please contact question.papers@sqa.org.uk.
- Question 5 [pim pic/Shutterstock.com](https://www.shutterstock.com/pim-pic/)